

Influencer Marketing ROI & Performance Analysis

Project Report

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Dataset: influencer_marketing_roi_dataset.csv (150,000 campaign records)

Tools: Python (pandas, NumPy, matplotlib, seaborn) in a Google Colab notebook

1. Executive Summary

This report analyzes 150,000 influencer marketing campaigns spanning four platforms (Instagram, YouTube, TikTok, Twitter) and seven influencer categories, to answer four questions: which platforms and categories deliver the best return on investment, how campaign types compare on performance, how engagement/reach/sales trends move over time, and which influencer categories and campaign types deserve priority in future partnerships.

Because the raw dataset does not include a spend or price field, ROI figures in this report are built on a transparent, adjustable cost model (Section 3) rather than actual financial data. All directional findings below should be re-validated once real campaign spend and revenue figures are available; the efficiency metrics that do not depend on cost assumptions (engagement rate, sales per 1,000 reach) are unaffected by this caveat.

Headline findings

- Twitter shows the strongest assumed ROI among platforms (~3,620% overall), driven mainly by its lower assumed cost-per-engagement rather than materially higher sales.
 - ROI differences across influencer categories are narrow (roughly 1,987%–2,015%) — no single category dominates on cost-adjusted return.
 - Fashion (composite score 77/100) and Tech (71/100) rank highest among influencer categories when ROI, conversion efficiency, and engagement rate are combined.
 - Seasonal Sale (75/100) is the strongest-scoring campaign type, followed by Giveaway (58/100); Product Launch scores lowest (17/100) despite reasonable average sales.
 - All ten of the highest-ROI reliable platform x category x campaign-type combinations involve Twitter, led by Twitter x Food x Product Launch (~20,945% ROI, n=421 campaigns).
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2. Objectives

- ROI Analysis — quantify return on investment for influencer campaigns across platforms and categories.
 - Campaign Benchmarking — compare campaign performance by type, influencer niche, and platform.
 - Trend Analysis — track engagement, reach, and sales trends over time.
 - Influencer Selection — identify high-performing influencer categories and campaign types for future partnerships.
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3. Data Overview & Methodology

3.1 Dataset structure

The dataset contains 150,000 unique campaign records with no missing values and no duplicate campaign IDs. Each row represents one campaign with the following fields:

Field	Description
campaign_id	Unique campaign identifier
platform	Instagram, YouTube, TikTok, or Twitter
influencer_category	Fashion, Food, Travel, Beauty, Tech, Gaming, or Fitness
campaign_type	Giveaway, Product Launch, Brand Awareness, Seasonal Sale, or Event Promotion
start_date / end_date	Campaign start and end timestamps
engagements	Total user engagements recorded
estimated_reach	Estimated audience reach
product_sales	Units sold attributed to the campaign
campaign_duration_days	Campaign length in days

Campaign volume is not evenly split by platform: Instagram accounts for the largest share (59,819 campaigns, ~40%), followed by YouTube (45,139, ~30%), TikTok (29,984, ~20%), and Twitter (15,058, ~10%). Influencer categories and campaign types are each close to evenly distributed.

3.2 Data quality note

The start_date field increases sequentially by one day per row, beginning 2022-01-01 and running through 2432-09-07 — i.e., dates were generated in sequence rather than reflecting real calendar scheduling. Trend results (Section 6) should therefore be read as “how metrics evolve across the campaign sequence” rather than genuine seasonal or year-over-year patterns. Engagement and reach also appear to be generated independently of one another (engagement rate ranges from under 1% to over 9,000% of reach for individual campaigns), which is not realistic for real-world social data and should be kept in mind when interpreting per-campaign efficiency figures; aggregated averages are more stable.

3.3 ROI methodology and assumptions

The dataset has no cost/spend or revenue field, so ROI cannot be computed directly. To make ROI calculable, this analysis applies an explicit, editable cost model:

Cost = engagements × assumed cost-per-engagement (CPE), which varies by platform

Revenue = product_sales × assumed average order value (AOV = \$45)

ROI % = (Revenue – Cost) / Cost × 100

Platform	Assumed CPE (USD)
Instagram	\$0.10
TikTok	\$0.08
YouTube	\$0.15
Twitter	\$0.06

These figures are illustrative, industry-style placeholders, not actual spend data. They are set as variables at the top of the accompanying notebook so real cost and pricing data can be substituted directly, after which every downstream table and chart recalculates automatically. Because Twitter and TikTok carry the lowest assumed CPE, they mechanically show the highest ROI in this report — that result reflects the cost assumption as much as campaign performance, which is why Section 4 also reports assumption-free efficiency metrics (engagement rate, sales per 1,000 reach) alongside ROI.

Three additional metrics were derived that do not depend on any cost assumption: engagement rate (engagements ÷ reach), sales per 1,000 reach (a conversion-efficiency proxy), and sales per engagement.

4. ROI Analysis

Table 1 summarizes assumed cost, revenue, and overall ROI by platform (“overall ROI” = total revenue vs. total cost across all campaigns on that platform, which is more robust than averaging each campaign's ROI %).

Platform	Campaigns	Avg. Cost/Sale (\$)	Overall ROI (%)
Twitter	15,058	5.37	3,621.5%
TikTok	29,984	7.05	2,714.9%
Instagram	59,819	8.83	2,140.3%
YouTube	45,139	13.75	1,400.5%

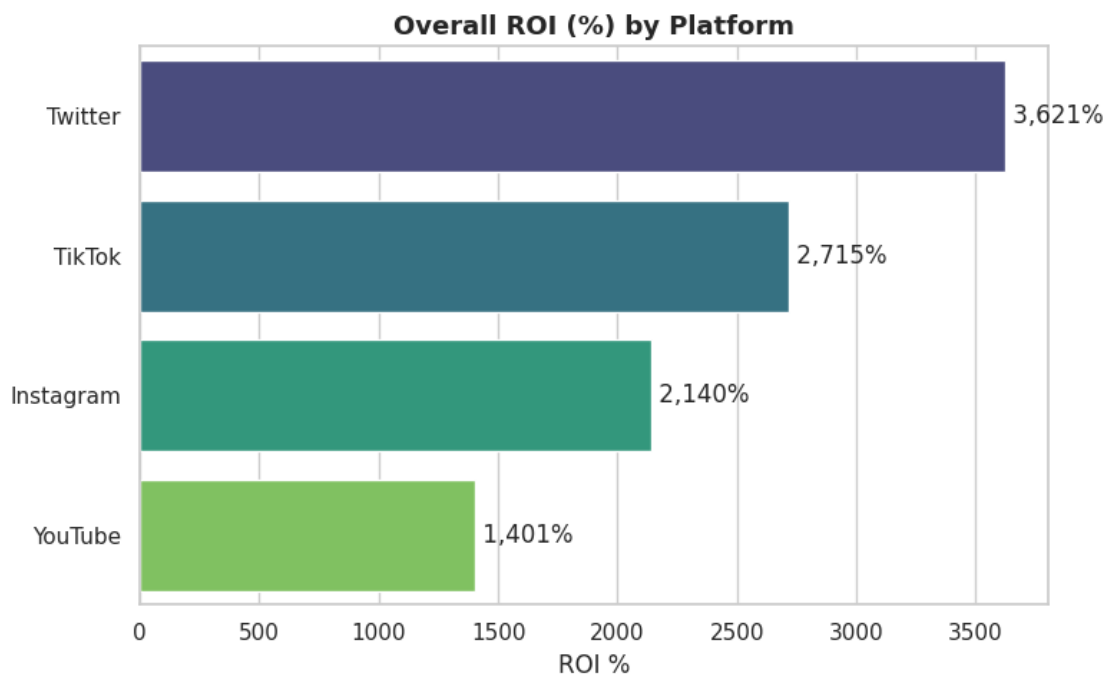


Figure 1. Overall ROI (%) by platform, under the assumed cost model.

By influencer category, ROI is far more tightly clustered — a roughly 30-point spread on a scale where the platform figures above range from 1,400% to 3,621%:

Category	Campaigns	Avg. Cost/Sale (\$)	Overall ROI (%)
Travel	21,344	10.31	2,014.8%
Fitness	21,211	9.82	2,011.2%
Tech	21,545	9.22	2,001.6%
Food	21,659	9.34	1,993.2%
Fashion	21,502	9.08	1,989.1%
Beauty	21,340	9.11	1,987.8%
Gaming	21,399	10.38	1,986.4%

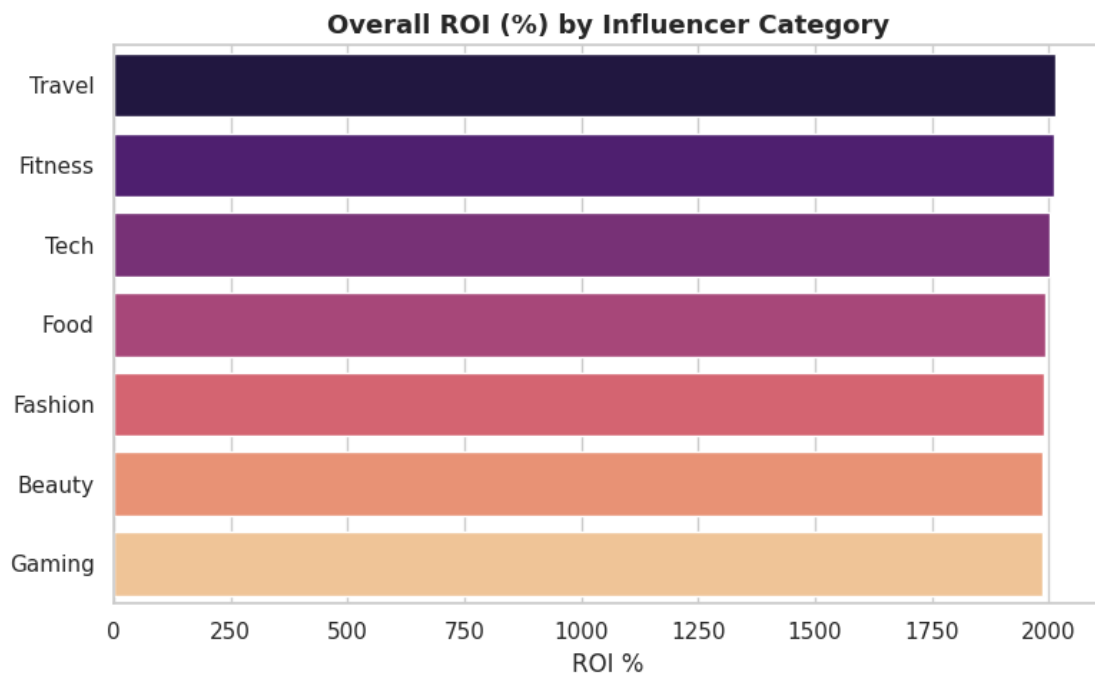


Figure 2. Overall ROI (%) by influencer category — note the narrow range on the x-axis.

The narrow spread indicates that, in this dataset, category choice alone is a weak lever for ROI once cost is factored in — platform selection and campaign type (below) matter more.

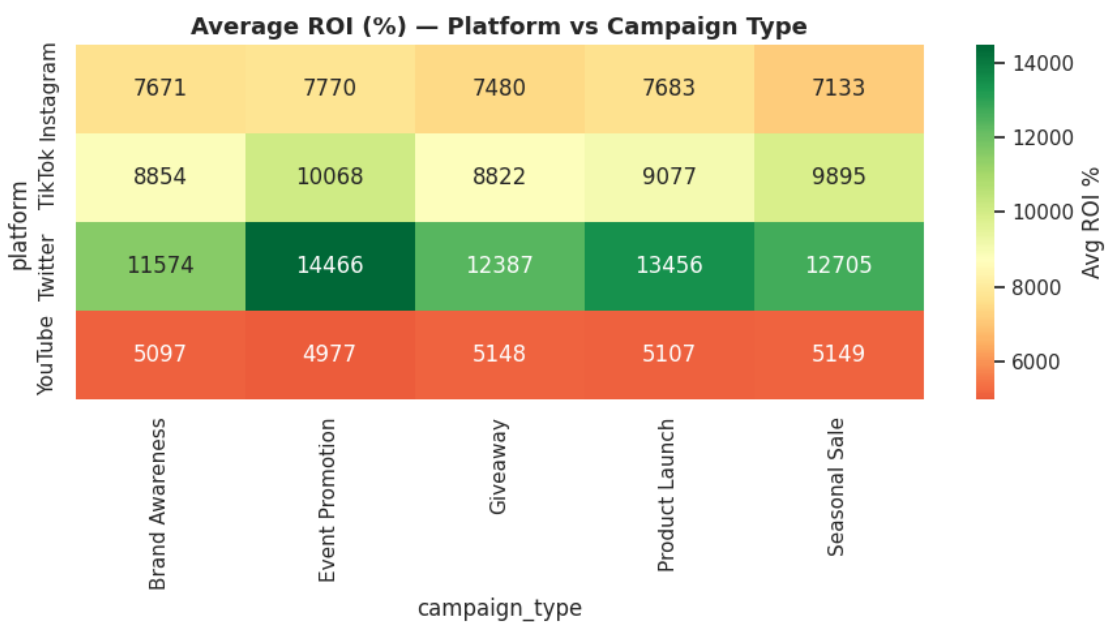


Figure 3. Average ROI (%) by platform and campaign type. Twitter rows are consistently the strongest performers; Event Promotion is the strongest campaign type on average.

5. Campaign Benchmarking

Table 3 compares campaign types on raw performance and efficiency, independent of the ROI cost assumptions.

Campaign Type	Avg. Sales	Engagement Rate (%)	Sales / 1,000 Reach	Avg. ROI (%)
Event Promotion	2,496.0	33.1	16.45	8,045.1%

Campaign Type	Avg. Sales	Engagement Rate (%)	Sales / 1,000 Reach	Avg. ROI (%)
Product Launch	2,493.0	33.2	16.34	7,773.2%
Seasonal Sale	2,499.4	34.8	17.26	7,656.8%
Giveaway	2,498.2	34.7	17.00	7,535.1%
Brand Awareness	2,502.2	34.1	16.97	7,525.3%

Average sales are nearly identical across campaign types (all within 1% of ~2,498 units), so differences in ROI and conversion efficiency are driven by how efficiently engagement converts, not by raw sales volume.

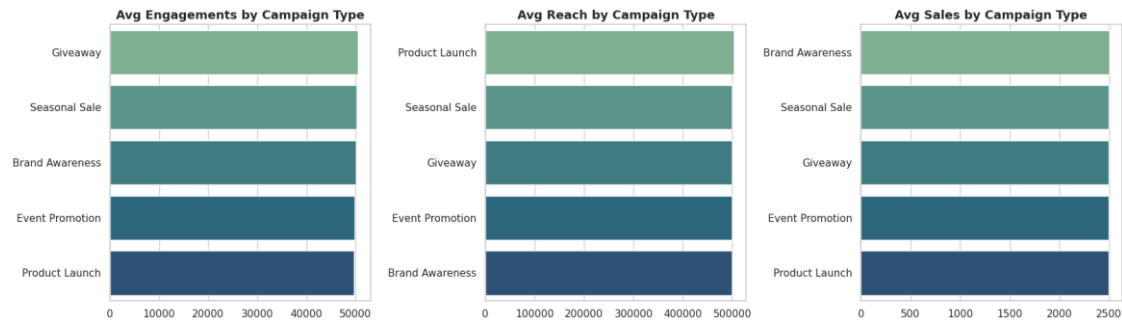


Figure 4. Average engagements, reach, and sales by campaign type.

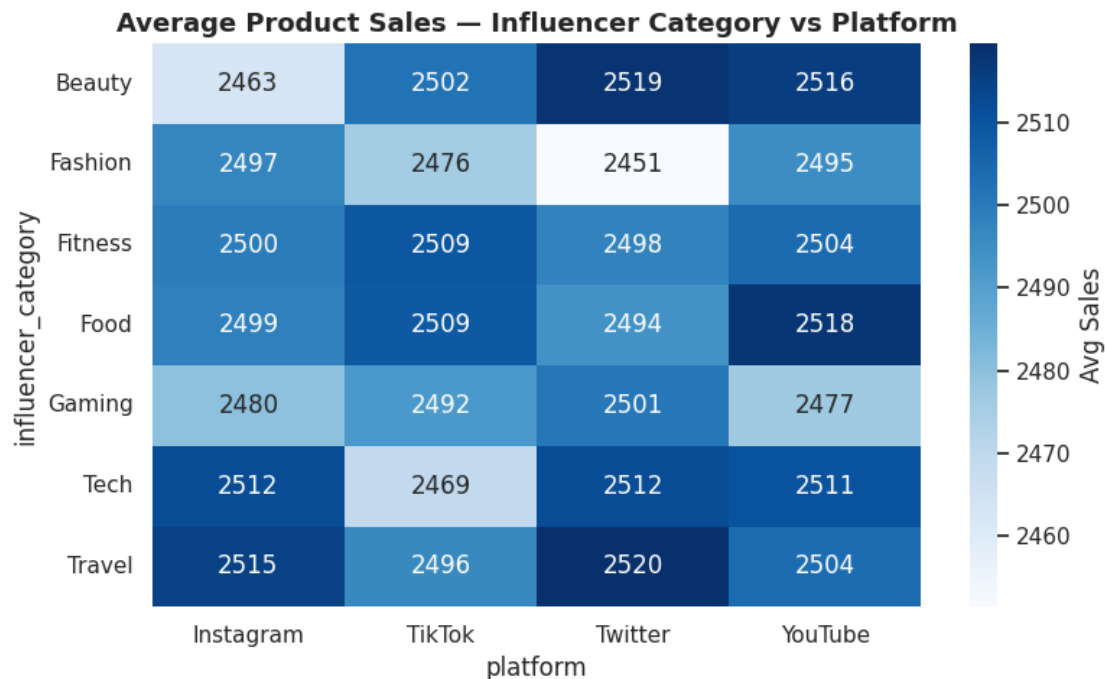


Figure 5. Average product sales — influencer category vs. platform.

6. Trend Analysis

Monthly-aggregated trends (with a 6-month rolling average) are shown below. As noted in Section 3.2, the date field increases sequentially rather than reflecting real scheduling, so these charts are best read as how performance evolves across the campaign sequence rather than true seasonality.

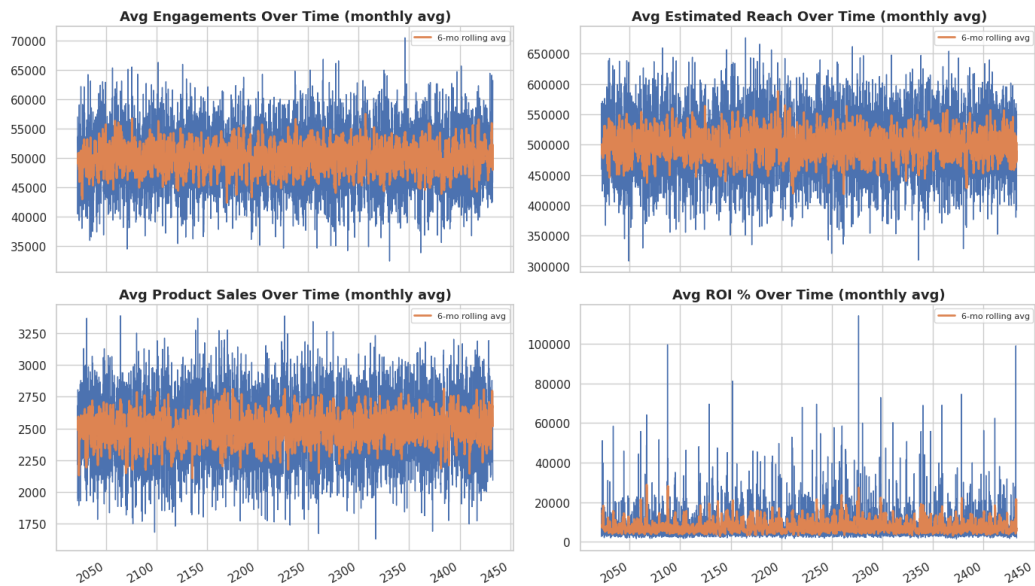


Figure 6. Monthly average engagements, reach, sales, and ROI, with 6-month rolling averages.

Engagement, reach, and sales all fluctuate within a stable band over time with no strong upward or downward drift — consistent with the underlying values being generated independently of the sequential date field rather than trending. ROI shows more volatility because it compounds engagement, sales, and platform-mix effects month to month.

7. Influencer Selection

To avoid over-weighting any single metric, ROI, sales-per-1,000-reach, and engagement rate were each min-max normalized to 0–100 and averaged into a composite score per influencer category and per campaign type.

Influencer Category	Avg. ROI (%)	Engagement Rate (%)	Sales / 1,000 Reach	Composite Score
Fashion	7,644.9%	35.65	17.69	77.0
Tech	7,937.8%	34.50	17.06	71.0
Travel	7,413.2%	36.02	17.19	56.6
Food	7,754.8%	33.46	16.68	42.6
Gaming	7,971.2%	32.60	16.24	38.6
Fitness	7,537.2%	33.09	16.71	26.8
Beauty	7,683.7%	32.48	16.04	16.2
Campaign Type	Avg. ROI (%)	Engagement Rate (%)	Sales / 1,000 Reach	Composite Score
Seasonal Sale	7,656.8%	34.75	17.26	75.1
Giveaway	7,535.1%	34.74	17.00	57.7
Brand Awareness	7,525.3%	34.13	16.97	43.7
Event Promotion	8,045.1%	33.09	16.45	37.3
Product Launch	7,773.2%	33.15	16.34	17.0

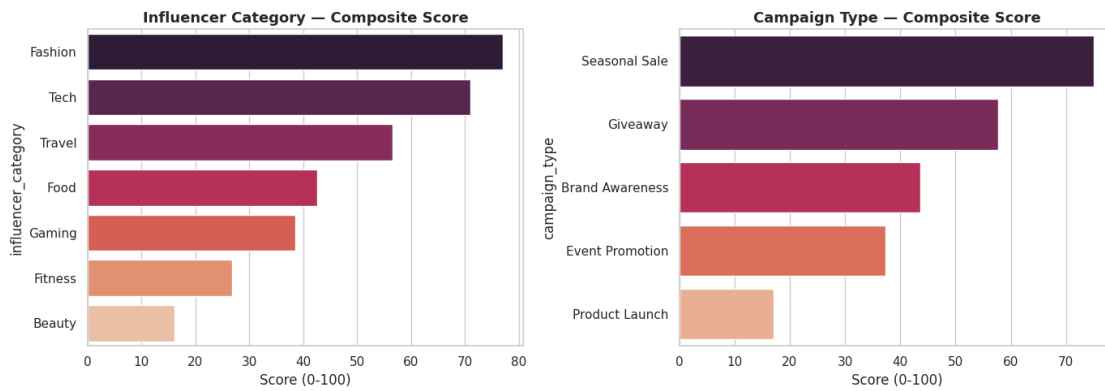


Figure 7. Composite performance score (0–100) by influencer category and campaign type.

Top reliable combinations

Filtering to platform × category × campaign-type combinations with at least 30 campaigns (for statistical reliability), the ten highest-ROI combinations were:

Platform	Category	Campaign Type	Avg. ROI (%)	n
Twitter	Food	Product Launch	20,944.5%	421
Twitter	Fitness	Event Promotion	20,668.6%	430
Twitter	Beauty	Event Promotion	17,433.7%	434
Twitter	Food	Brand Awareness	16,437.8%	441
Twitter	Gaming	Giveaway	15,504.9%	411
Twitter	Fitness	Giveaway	15,408.6%	410
Twitter	Tech	Event Promotion	15,343.9%	442
Twitter	Beauty	Product Launch	15,216.0%	423
Twitter	Fashion	Seasonal Sale	14,908.8%	456
Twitter	Fashion	Event Promotion	14,657.8%	464

All ten top combinations involve Twitter, again reflecting its lower assumed cost-per-engagement. With real spend data, this list should be re-run to confirm whether Twitter still leads or whether the ranking is primarily an artifact of the CPE assumption.

8. Recommendations

- Prioritize Fashion and Tech influencer categories, and Seasonal Sale / Giveaway campaign types, which rank highest on the combined ROI + efficiency + engagement score.
- Treat platform-level ROI rankings with caution until real spend data replaces the assumed CPE figures — the current Twitter/TikTok advantage is partly a function of the cost assumption, not necessarily superior campaign execution.
- Because average sales are nearly flat across campaign types (Section 5), focus optimization on conversion efficiency (sales per 1,000 reach, engagement rate) rather than assuming any one campaign format inherently drives more volume.
- Validate the ten top-ROI platform x category x campaign-type combinations (Section 7) with actual spend and revenue once available, since they currently cluster entirely on one platform.
- Re-run the trend analysis once real, non-sequential campaign dates are available to capture genuine seasonal patterns.

9. Limitations

- No actual cost/spend or revenue data was available; ROI figures rely on assumed CPE and AOV constants and should be treated as directional, not decision-grade, until replaced with real figures.
 - The date field is sequential rather than reflecting real campaign scheduling, limiting the reliability of seasonal trend conclusions.
 - Engagement and reach values appear generated independently rather than realistically correlated, which inflates the variance of per-campaign efficiency ratios (aggregated/group-level figures are more reliable than single-campaign figures).
 - The analysis is descriptive/correlational; it does not establish that a given platform, category, or campaign type causes higher sales.
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10. Appendix

10.1 Tools and reproducibility

All analysis was performed in Python (pandas, NumPy, matplotlib, seaborn) inside a Google Colab notebook (influencer_marketing_roi_analysis.ipynb), which reproduces every table and figure in this report and exposes the CPE/AOV assumptions as editable variables.

10.2 Dataset summary statistics

Metric	Mean	Median
Engagements	50,065	50,101
Estimated reach	500,240	500,074
Product sales	2,498	2,501
Campaign duration (days)	15.0	15